

# Forensic lenses — math cheat sheet

ADAIS / wave\_1      one slide per lens

## NLF — Noise-Level Function

For every  $8 \times 8$  patch  $p$  of the gray image, compute mean and std:

$$\bar{I}_p = \frac{1}{64} \sum_{(i,j) \in p} I(i,j), \quad \sigma_p = \sqrt{\frac{1}{64} \sum_{(i,j) \in p} (I(i,j) - \bar{I}_p)^2}.$$

Throw away the most textured 50% of patches (high local gradient). Bin the rest by  $\bar{I}_p$  into 16 intensity bins; for each bin take the median  $\sigma$ . Fit the **Poisson–Gaussian sensor model**:

$$\boxed{\sigma^2(I) = a \cdot I + b} \quad R^2 = 1 - \frac{\sum_i (\sigma_i^2 - \hat{\sigma}_i^2)^2}{\sum_i (\sigma_i^2 - \bar{\sigma}^2)^2}.$$

**Output (20 values):** the 16 bin medians (the curve) plus  $(a, b, R^2, \text{valid\_frac})$ .

*What to say: Real cameras have shot noise (Poisson) that grows with brightness, plus a constant read noise (Gaussian). The line should fit. AI generators don't simulate sensor physics, so the curve is flat or noisy and  $R^2$  is low.*

## CFA — Colour-Filter-Array periodicity

For each channel  $c \in \{R, G, B\}$ , take the high-pass residual (subtract a Gaussian blur with  $\sigma=1$ ):

$$r_c = c - G_\sigma * c.$$

Take the 2-D FFT magnitude (shifted so DC is at the centre):

$$S_c(\omega_y, \omega_x) = |\mathcal{F}\{r_c\}|.$$

Sample three diagnostic peaks at the half-Nyquist frequencies and normalise by the mean spectrum so amplitude doesn't matter:

$$\tilde{p}_{c,k} = \frac{\max_{\|(\omega_y, \omega_x) - q_k\| \leq 2} S_c(\omega_y, \omega_x)}{\bar{S}_c}, \quad q_k \in \left\{ \left(0, \frac{\pi}{2}\right), \left(\frac{\pi}{2}, 0\right), \left(\frac{\pi}{2}, \frac{\pi}{2}\right) \right\}.$$

**Output (12 values):** 3 channels  $\times$  {hh, vh, dh, pool}.

*What to say: Cameras only sense one colour per pixel (Bayer mosaic) and interpolate the missing two from neighbours. That interpolation leaves a 2-pixel periodicity, which shows up as a spike at  $\pi/2$  in the FFT. Generators emit RGB directly — no spike.*

## RES (radial) — residual spectrum, radial bins

Take the denoising residual of the gray image  $r = I - D_{\text{wavelet}}(I)$ , then its FFT magnitude:

$$S(\omega_y, \omega_x) = |\mathcal{F}\{r\}|, \quad \rho(\omega_y, \omega_x) = \sqrt{(\omega_y - c_y)^2 + (\omega_x - c_x)^2}.$$

Define 64 *log-spaced* radius bins  $1 = \rho_0 < \rho_1 < \dots < \rho_{64} = N/2 - 1$ . The feature for bin  $k$

is the mean spectrum on that ring, L1-normalised:

$$P_k = \frac{1}{|B_k|} \sum_{(\omega_y, \omega_x) \in B_k} S(\omega_y, \omega_x), \quad \tilde{P}_k = \frac{P_k}{\sum_j P_j},$$

$$B_k = \{(\omega_y, \omega_x) : \rho_k \leq \rho(\omega_y, \omega_x) < \rho_{k+1}\}.$$

**Output:** 64 values — the *shape* of the radial spectrum.

*What to say: Real photos have a smooth  $1/f$  fall-off. GAN and diffusion upsamplers leave periodic copies of the spectrum — visible as bumps at specific radii. The radial profile is the cleanest way to spot them.*

## RES (angular) — residual spectrum, angular bins

Same residual  $r$  and spectrum  $S$  as above, but now bin by orientation inside a mid-frequency annulus:

$$\theta(\omega_y, \omega_x) = \text{atan2}(\omega_y - c_y, \omega_x - c_x) \bmod \pi \quad (\text{fold to } [0, \pi)).$$

Mask to the annulus  $\mathcal{A} = \{0.25 r_{\text{nyq}} \leq \rho \leq 0.75 r_{\text{nyq}}\}$  and bin into 16 sectors of  $11.25^\circ$ :

$$Q_k = \frac{1}{|\mathcal{A}_k|} \sum_{(\omega_y, \omega_x) \in \mathcal{A}_k} S(\omega_y, \omega_x), \quad \tilde{Q}_k = \frac{Q_k}{\sum_j Q_j},$$

$$\mathcal{A}_k = \mathcal{A} \cap \{\theta \in [\theta_k, \theta_{k+1})\}.$$

**Output:** 16 values — the orientation pattern of the residual.

*What to say:* A real photo is roughly isotropic in the mid-frequencies — all directions look the same. Each generator’s upsampling kernel has a preferred axis (horizontal, vertical,  $45^\circ$  lobes), so the angular profile is what tells LlamaGen vs. RAR vs. VAR apart.

## LCA — Lateral Chromatic Aberration

Split the image into a  $4 \times 4$  grid of tiles. For each tile, align the R and B channels to G by integer cross-correlation (within  $\pm 3$  px):

$$(\delta y, \delta x)^* = \underset{|\delta y|, |\delta x| \leq 3}{\operatorname{argmax}} \frac{\sum (c_\delta - \bar{c})(G - \bar{G})}{\|c_\delta - \bar{c}\| \|G - \bar{G}\|} \quad (\text{normalised cross-correlation}).$$

Each tile centre has position  $(y_t, x_t)$  relative to the image centre, distance  $d_t = \sqrt{y_t^2 + x_t^2}$ , outward unit vector  $\hat{n}_t$ . Decompose the shift into radial / tangential components:

$$\Delta_{\text{rad},t} = (\delta y, \delta x) \cdot \hat{n}_t, \quad \Delta_{\text{tan},t} = (\delta y, \delta x) \cdot \hat{n}_t^\perp.$$

Fit a line through the origin (LCA is zero at the optical centre):

$$\hat{s} = \frac{\sum_t \Delta_{\text{rad},t} d_t}{\sum_t d_t^2}, \quad R^2 = 1 - \frac{\sum_t (\Delta_{\text{rad},t} - \hat{s} d_t)^2}{\sum_t (\Delta_{\text{rad},t} - \overline{\Delta_{\text{rad}}})^2}.$$

**Output (8 values):** two pairs (R→G, B→G)  $\times (\hat{s}, R^2, |\overline{\Delta_{\text{tan}}}|, \text{std}(\Delta_{\text{rad}}))$ .

*What to say:* Glass refracts red and blue light by different angles, so a real lens shifts the R and B channels outward from the optical centre, with the shift growing linearly with distance. Real → high slope, high  $R^2$ , near-zero tangential. AI → random scatter,  $R^2 \approx 0$ .

### The one-sentence summary for the audience:

*“Every formula here measures something a real camera has that a generative model doesn’t simulate: sensor noise statistics, demosaicing periodicity, upsampling-free spectrum, and lens-glass chromatic dispersion.”*